SARA

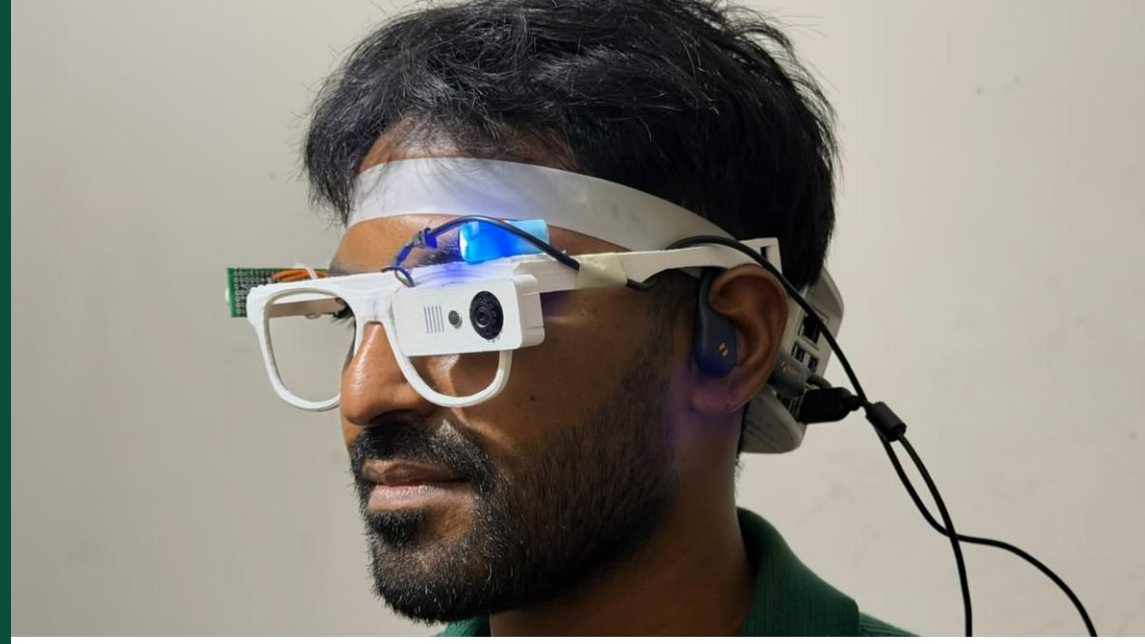
Smart Assistive Reading Aid: an offline smart glass that reads Bangla and English, names objects, and recognises and checks Bangladeshi Taka notes

MD Abdun Fattah Lam (2220014) · Mukut Protim Memo (2020086)

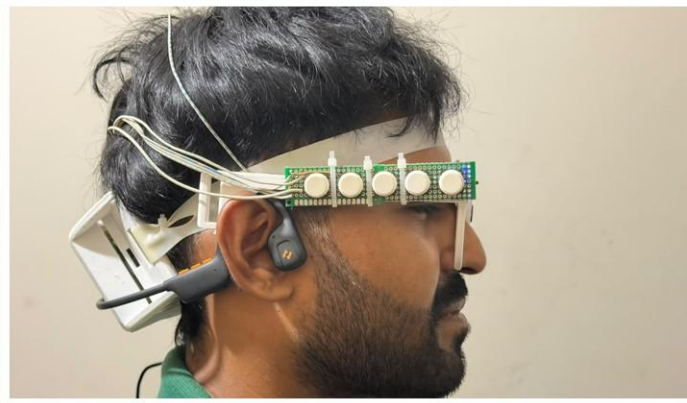
Tanjila Hossain Tapti (2130561)

Supervisor: Mohammad Shidujaman, PhD, Assistant Professor

Department of Computer Science and Engineering, Independent University, Bangladesh









Three everyday challenges

Money



Which note is this? Is it real?

Reading



What does this label say?

Surroundings



What is in front of me?

Today, every answer depends on another person, and it cannot be checked.

Why existing solutions fall short

2.2 billion

people worldwide have a near or distance vision impairment [1]

most of them in low- and middle-income regions [1]



Limited Bangla support

Reading glasses are built for English [2]



Internet dependence

Research glasses send pictures to a server [3, 4]



Limited Taka verification

No wearable names Taka notes or checks them safely

Sources: [1] WHO 2023; [2] OrCam MyEye; [3] Ali A. et al., IEEE Access 2021; [4] Brilli et al., arXiv 2024.

SARA in action: ask the glass, not a stranger



Our prototype

Speak
"Glass, check money"

Detect offline
Raspberry Pi 5, no internet

Hear the answer
Bangla speech + vibration



Money

Names the note; checks it safely



Reading

Reads the label aloud



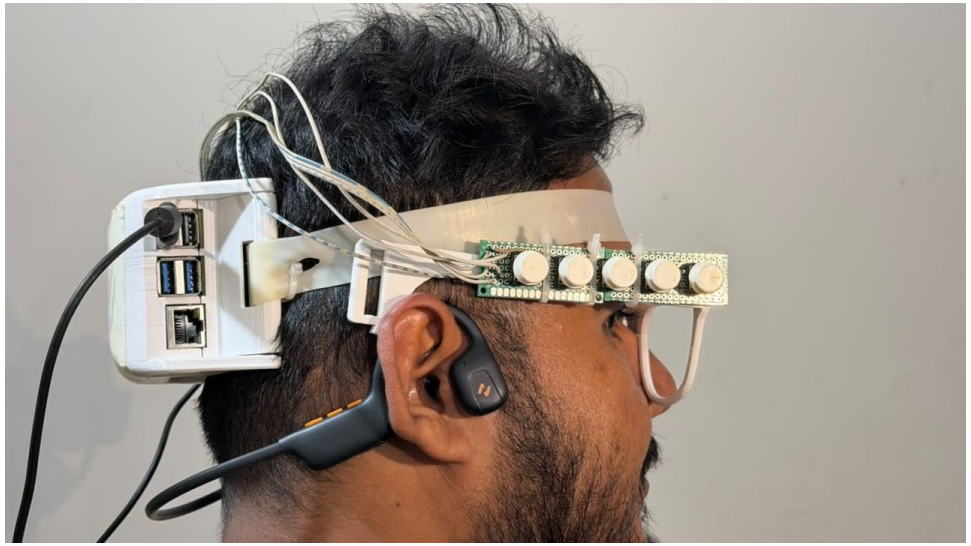
Surroundings

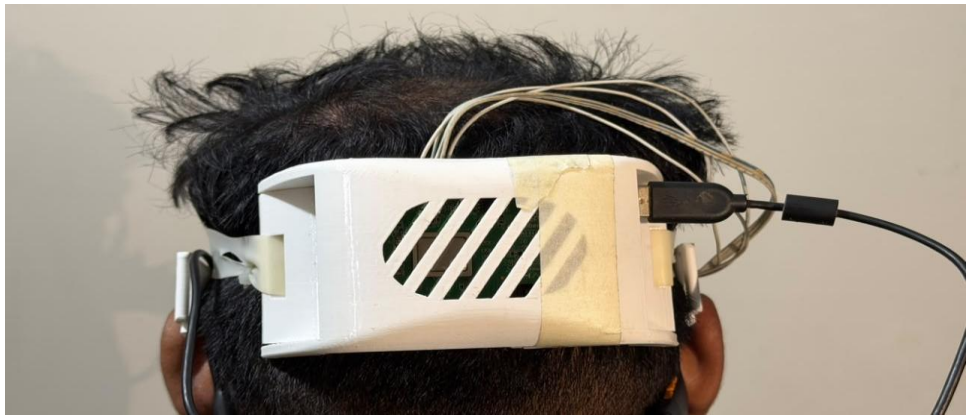
Names what is ahead

পাঁচশত টাকার নোট "500 Taka note": the answer the glass speaks

Illustrations except the prototype photo

Five modes, started by button or by voice





Text reading

Bangla and English

Objects

Named in Bangla

Taka notes

Value + safe check

Emotion

Face in front

Online description

Optional; needs internet

How the user starts a mode: five buttons on the frame, or a voice command: "Glass, read text"

Six research questions

1 System

Can it all run on a Raspberry Pi 5?

2 Synthetic data

Can unlabelled training data detect real notes?

3 Number of views

Can one view of a note be enough?

4 Unseen prints

Does the check hold on new counterfeit prints?

5 Safety

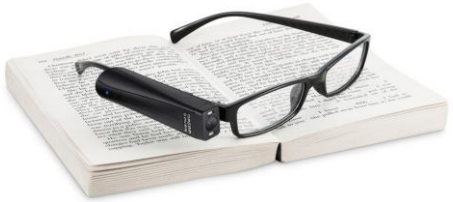
Can it never accuse a genuine note?

6 Bangla OCR

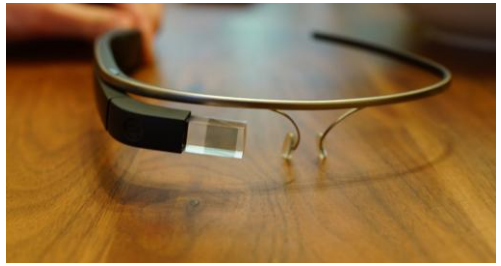
What matters most for reading Bangla?

What existing devices miss

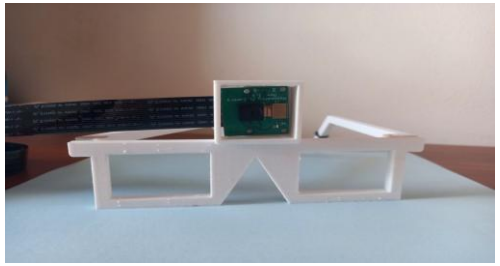
System	Reads Bangla	Knows Taka	Works offline	Wearable
OrCam MyEye [2]	No	No	Yes	Yes
Google Glass scene analysis [3]	No	No	No	Yes
AIris (2024) [4]	No	No	No	Yes
Raspberry Pi reading aids	No	No	Partly	Partly
SARA (ours)	**Yes**	**Yes**	**Yes**	**Yes**



OrCam MyEye [2]



Google Glass analysis [3]



AIris [4]

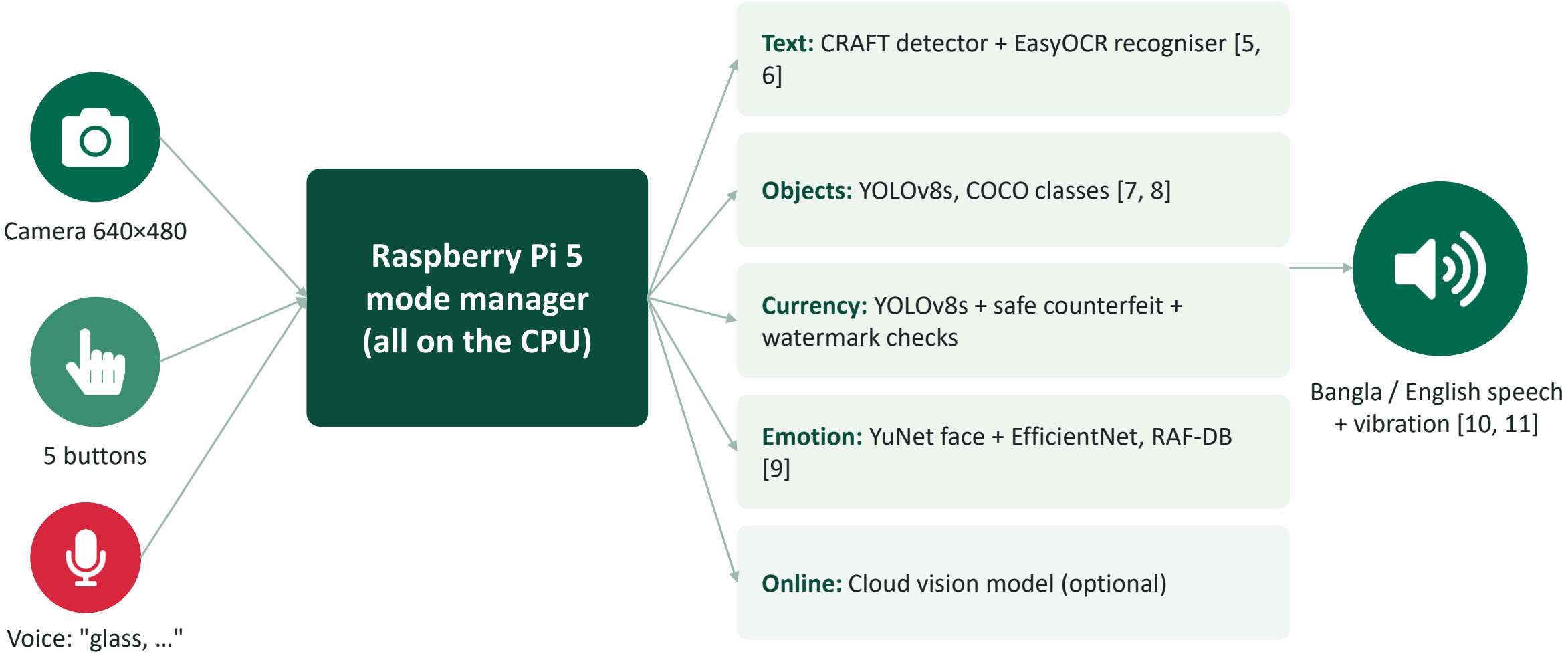
The research gap

No wearable combines Bangla, Taka and offline use.

- Counterfeit tests reuse the same prints
- Few report speed on the real device

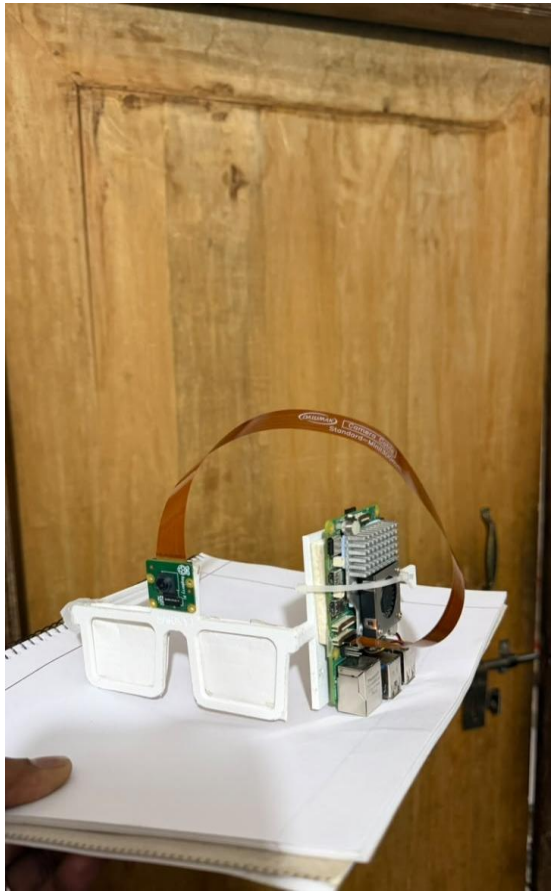
Photos: OrCam MyEye by Ylip, CC BY-SA 4.0; Google Glass by Ted Eytan, CC BY-SA 2.0 (Wikimedia Commons); AIris from [4]. Comparison from [2, 3, 4].

How the glass works



Sources: [5] Baek et al. 2019; [6] EasyOCR; [7] Ultralytics YOLOv8; [8] COCO; [9] RAF-DB; [10] Piper; [11] eSpeak NG.

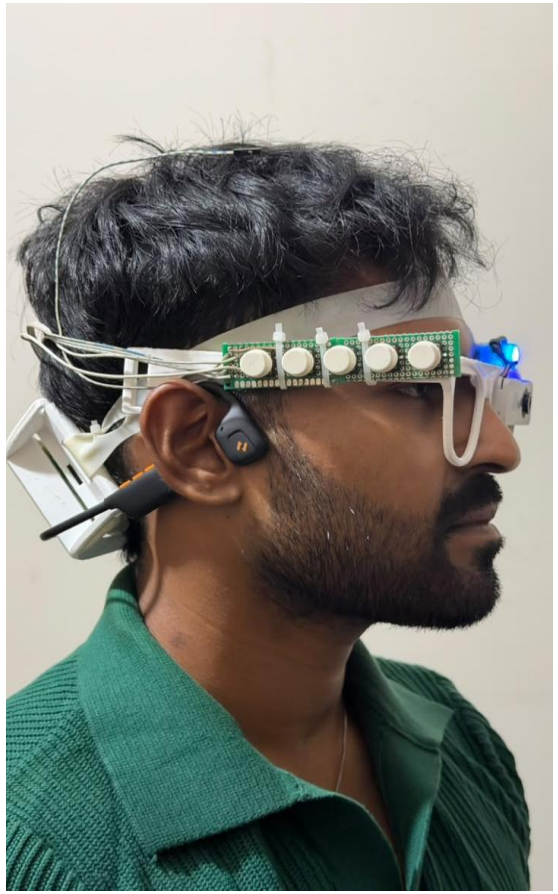
From a desk prototype to a wearable glass



Version 1
Board on one temple



Front
Camera + UV lamp



Side
5 buttons, bone-conduction earphone



Rear
Pi 5 in a ventilated case

Taka detection: 22,333 training images, no hand labels



Note photos [12] are pasted onto real scenes [8], and the program writes every label itself.

0.995

mAP@0.5 on 2,282 held-out composites

94.1%

right on 1,536 photos taken by other people

0

hand-drawn boxes needed

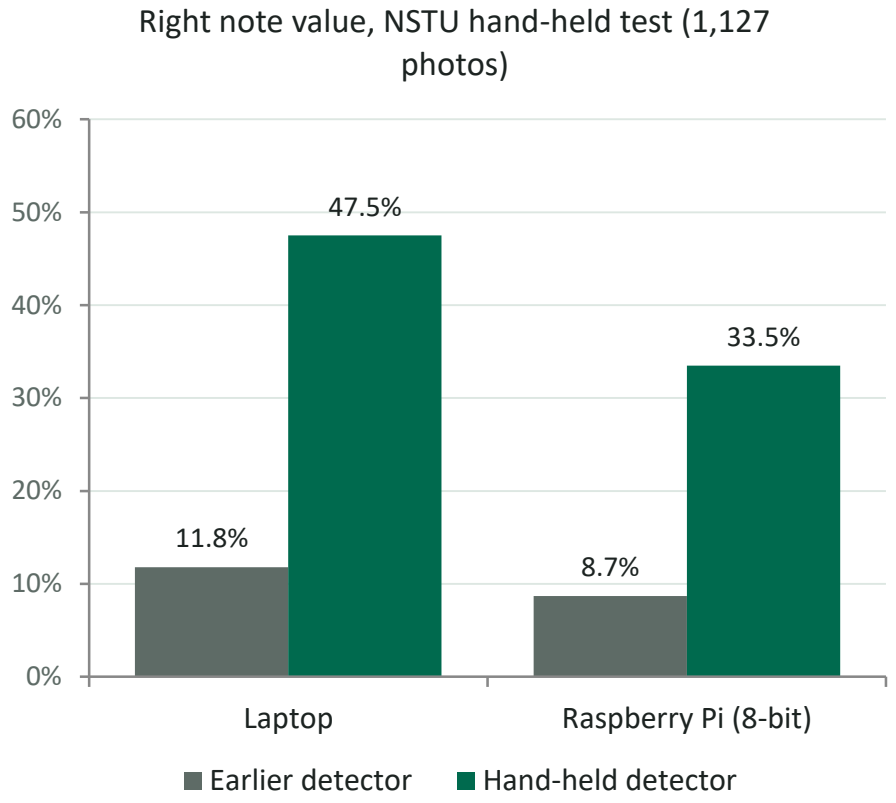
Sources: [12] BanglaTaka (Mendeley); [8] Lin et al., COCO, ECCV 2014; detector [7] Ultralytics YOLOv8. 94.1% with the hand-held detector (next slide).

Notes held in a hand: from 1 in 8 to 1 in 2



6,000 synthetic close-ups: folded, blurred, under fingers

Chosen on validation; the test set was read once

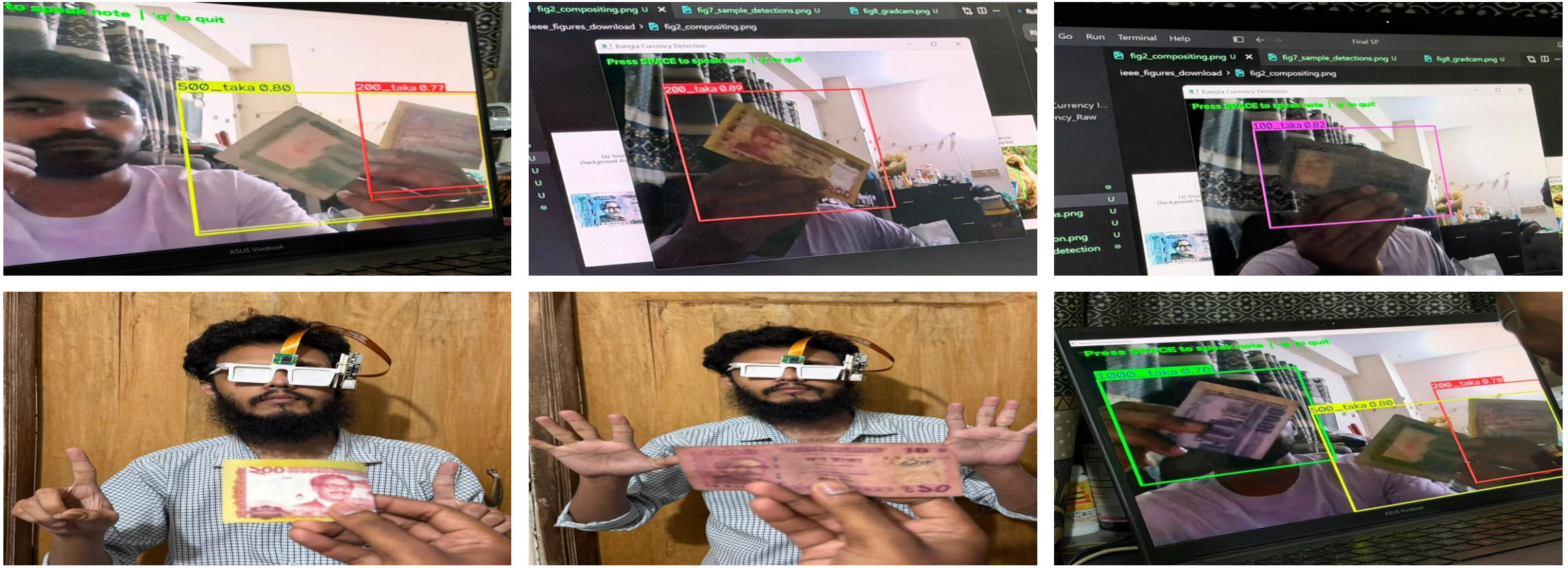


Trade-off: with 3 notes in view, the total is right 97.0% of the time (was 99.2%).

Sources: test photos [13] Rafi et al., NSTU-BDTAKA, Data in Brief 2024; detector [7] YOLOv8. Our results: paper_evidence/HANDHELD_FINETUNE.md.

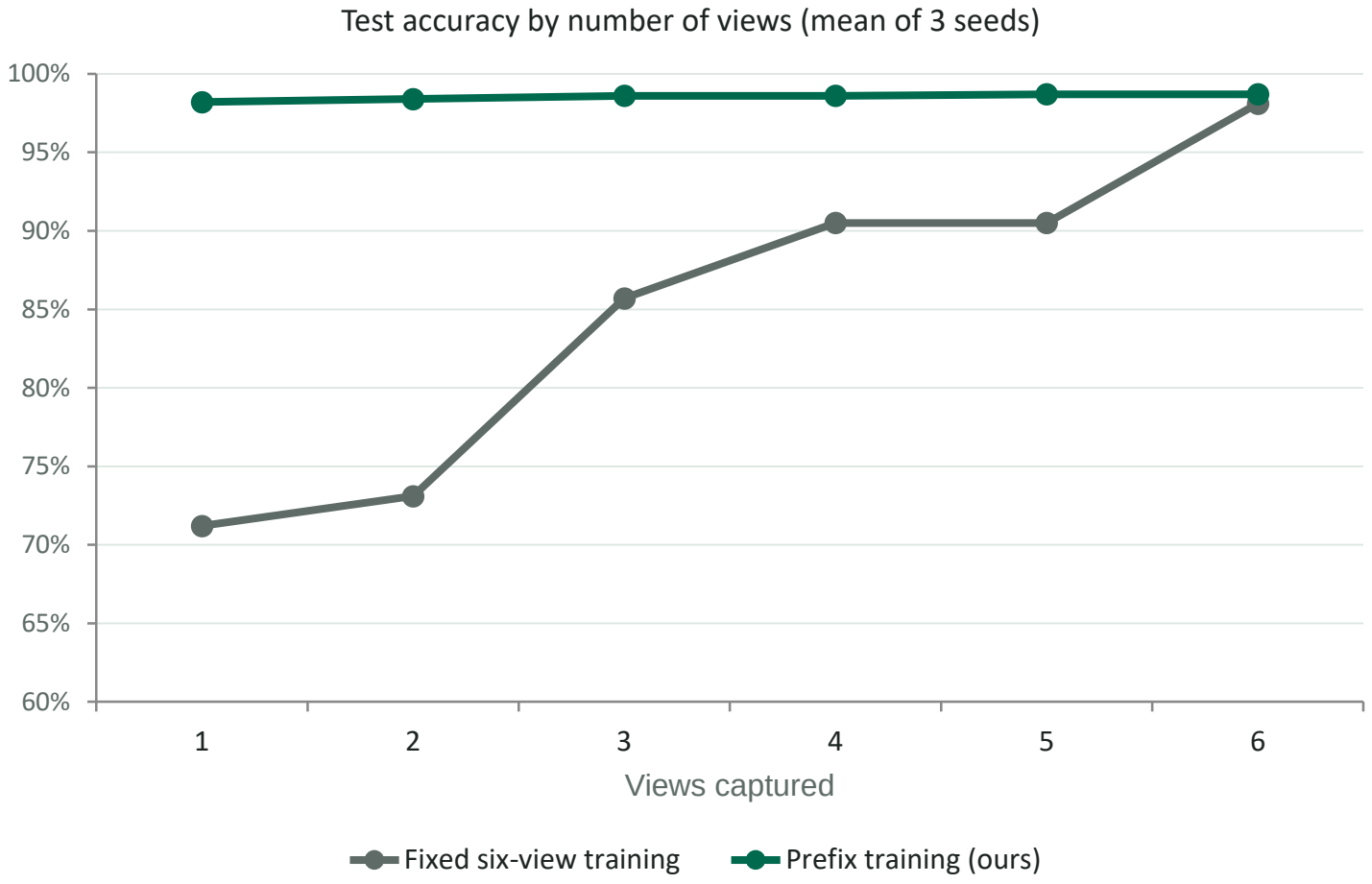
Live tests with real notes

Informal demonstrations, not a measured accuracy study



Six note values (10 to 1,000 Taka) named at 0.63–0.89 confidence, including worn and folded notes. Laptop webcam and the first glass prototype.

Counterfeit check: one view is enough with prefix training



With a single view

71.2%

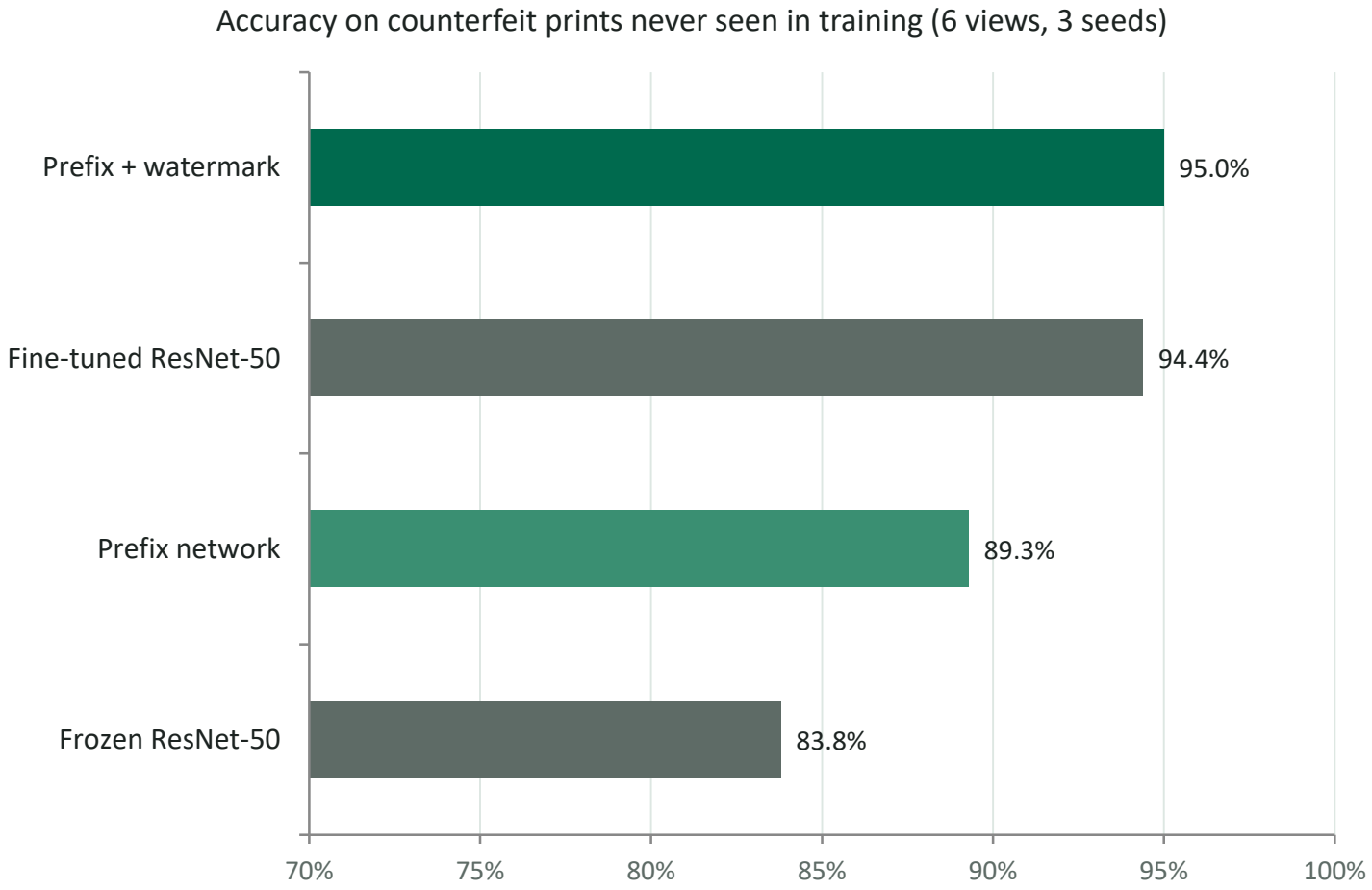
rises to

98.2%

Training on random first views: one view is enough.

Data: [14, 15] Piyas et al., JaalTaka dataset and article, 2025-2026 (1,390 notes, six views each; counterfeits seized by the Rapid Action Battalion).

Unseen counterfeit prints: a stricter test



279 of 322

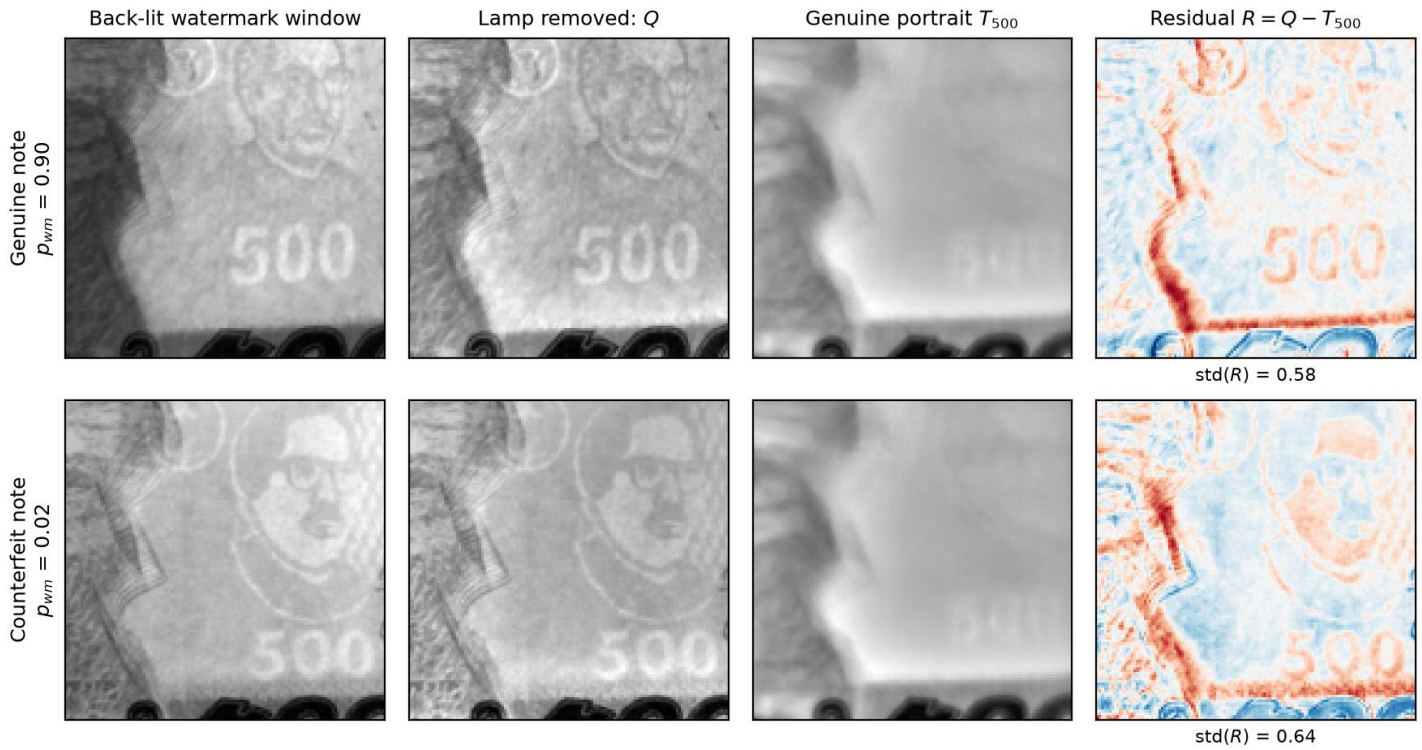
readable 500 Taka counterfeits carry the same serial number

Our split: every test counterfeit is a new print.

The watermark check lifts accuracy to 95.0%.

Data: [14] JaalTaka; serial-number audit and print-disjoint split are ours. Earlier Taka counterfeit work: [16] SVM on security features; [17] CNNs; [19] UV flecks + RPN.

DMWR: a tiny, physics-based watermark checker



1. Remove the lamp: Q = I / (G∗I)
2. Genuine portrait: T = median of Q over genuine notes
3. Residual R = Q − T → two small CNN streams → "watermark clear?"

91.5%

on unseen counterfeit prints (3 seeds, ±0.3)

47,818

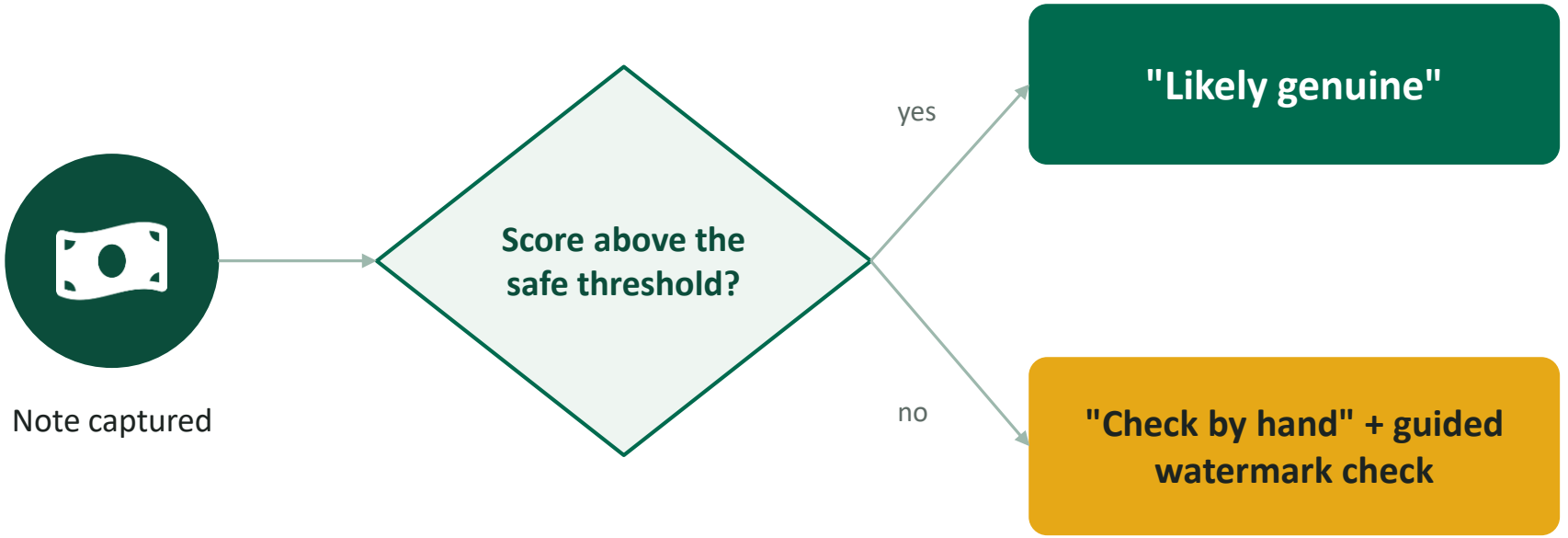
parameters (0.21 MB): 46× smaller than MobileNetV2 [18]

p ≥ 0.375

no significant difference from MobileNetV2 (92.9%)

Baseline [18] Sandler et al., MobileNetV2, CVPR 2018; data [14] JaalTaka. DMWR (denomination-matched watermark residual) is our method.

Safe before decisive: the glass never says "counterfeit"



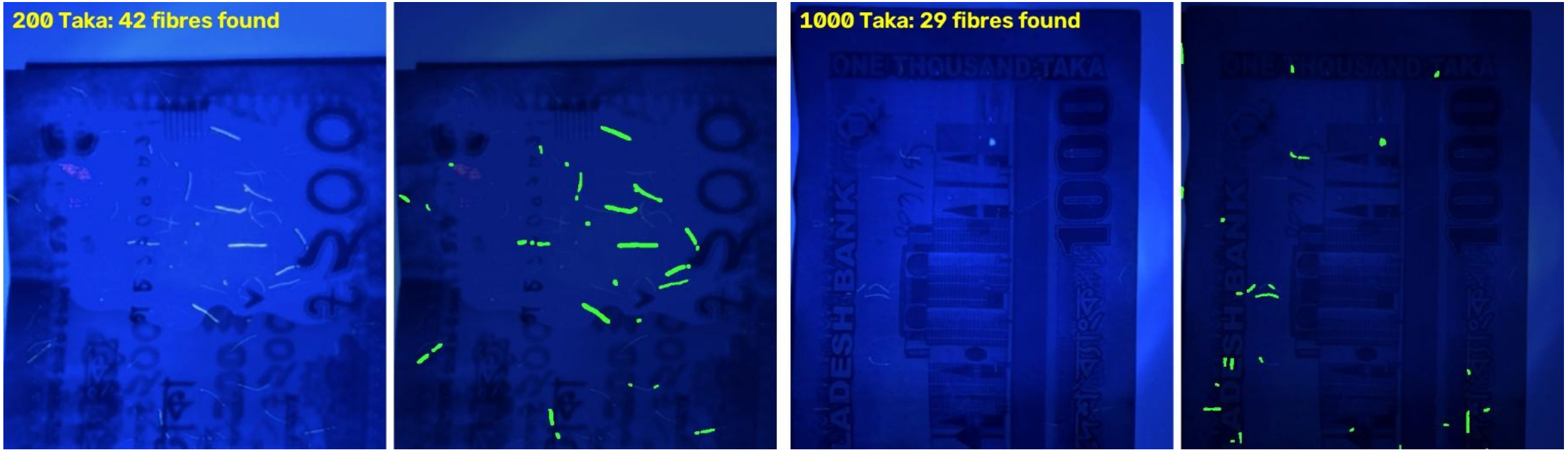
0 / 88

test counterfeits passed as "likely genuine"

The threshold is the highest score of any validation counterfeit. 57.5% of genuine notes are confirmed; for the rest, the glass asks for a check by hand.

Test notes: [14] JaalTaka.

UV fibre check: the glass carries its own UV lamp



85.6%

of 111 real UV photos of genuine notes accepted

88.6%

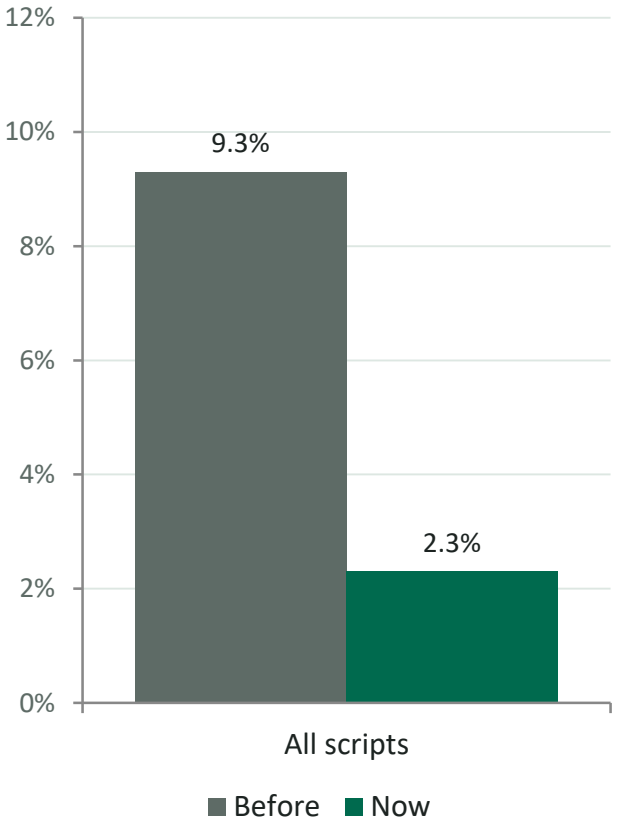
of simulated fibre-free printed sheets rejected

Prototype, off by default
Must still be validated on real counterfeit notes

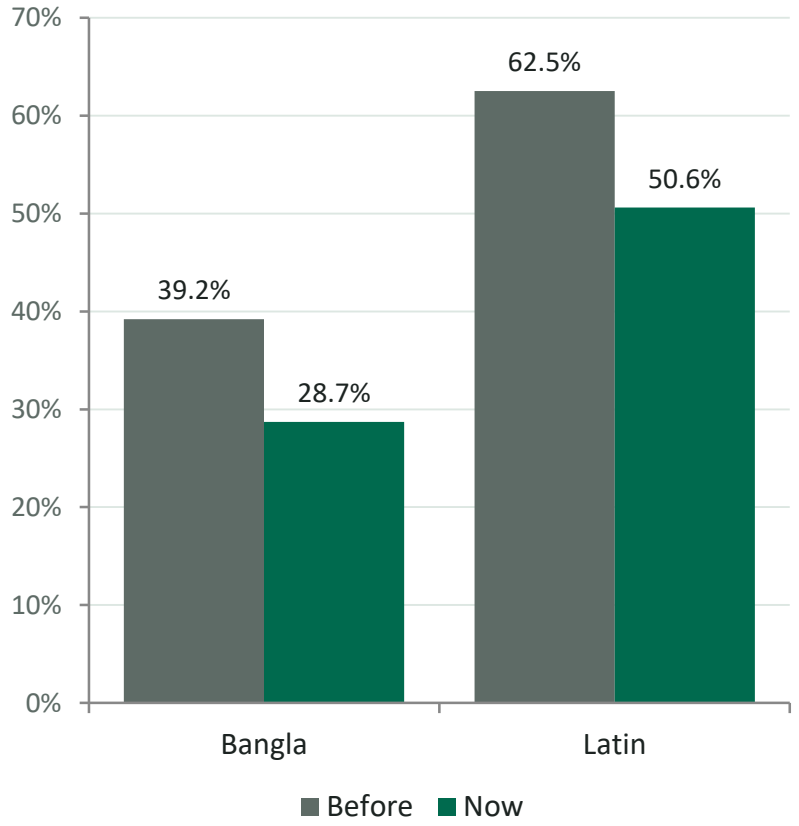
The glass's lamp emits 360-395 nm, covering the 365 nm used for banknote checks; banknote paper stays dark while its fibres glow [20]. Prior Taka UV work: [19] (96.3%). Fibre check and photos are ours.

Bangla reading: rendered text and real street photos

Rendered text (clean)



Real word photos (hard)



Character error rate, lower is better

What helped

- Reading the main text block only
- Tuned detector and clean-up
- Recogniser fine-tuned on real words

What did not

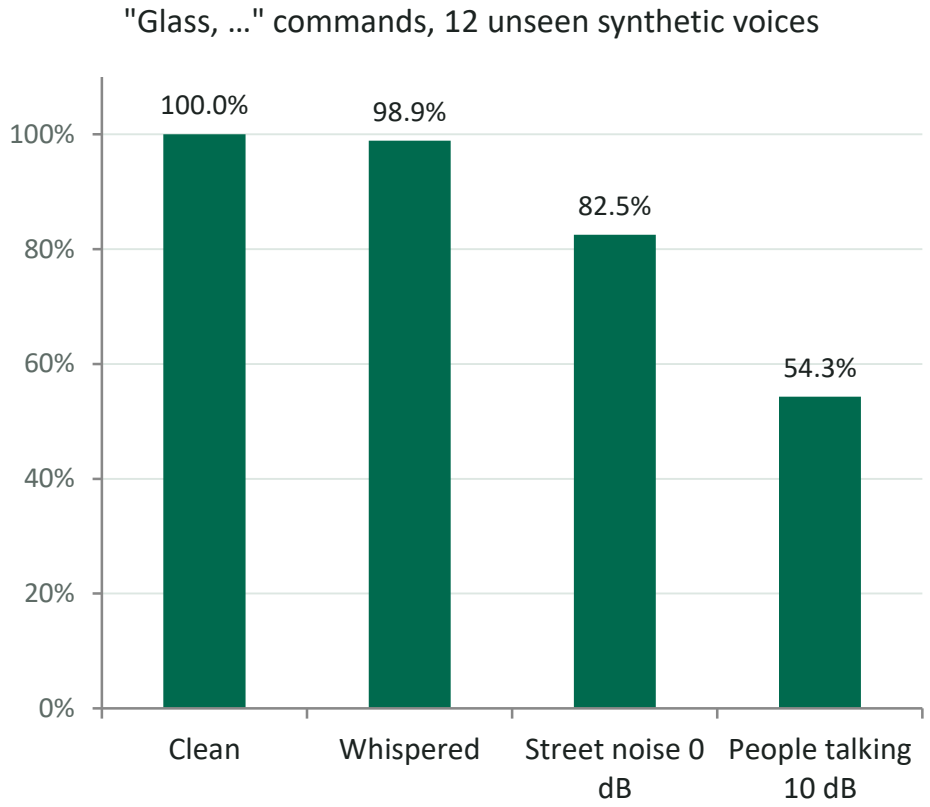
Binarisation, Tesseract, TrOCR, PaddleOCR, dictionaries

Engine [6] EasyOCR with [5] CRAFT; real words from [21] ICDAR 2019 MLT (1,000 test words). Every choice made on validation data.

Voice: a natural Bangla voice and spoken commands

Speaking

- Natural Piper voice, Bangla and English [10, 11]
- No broken sound when the Pi is busy
- "পড়ছি" (reading) while text is captured



212 → 13

false commands in 5,040 ordinary sentences, thanks to the wake word

0.86 s

from the end of speech to the action (median)

Command results: synthetic voices on a laptop, not human trials on the glasses

Recogniser [22] Vosk; noise from [23] Piczak, ESC-50; voices [10, 11]. Results: paper_evidence/VOICE_COMMANDS_FAST.md.

Running on the Raspberry Pi 5

399 ms
currency check, end to end

64 ms
watermark check

562 ms
object scan (YOLOv8s)

680 ms
emotion, face in view

18.3 s
3-line sign, text found once

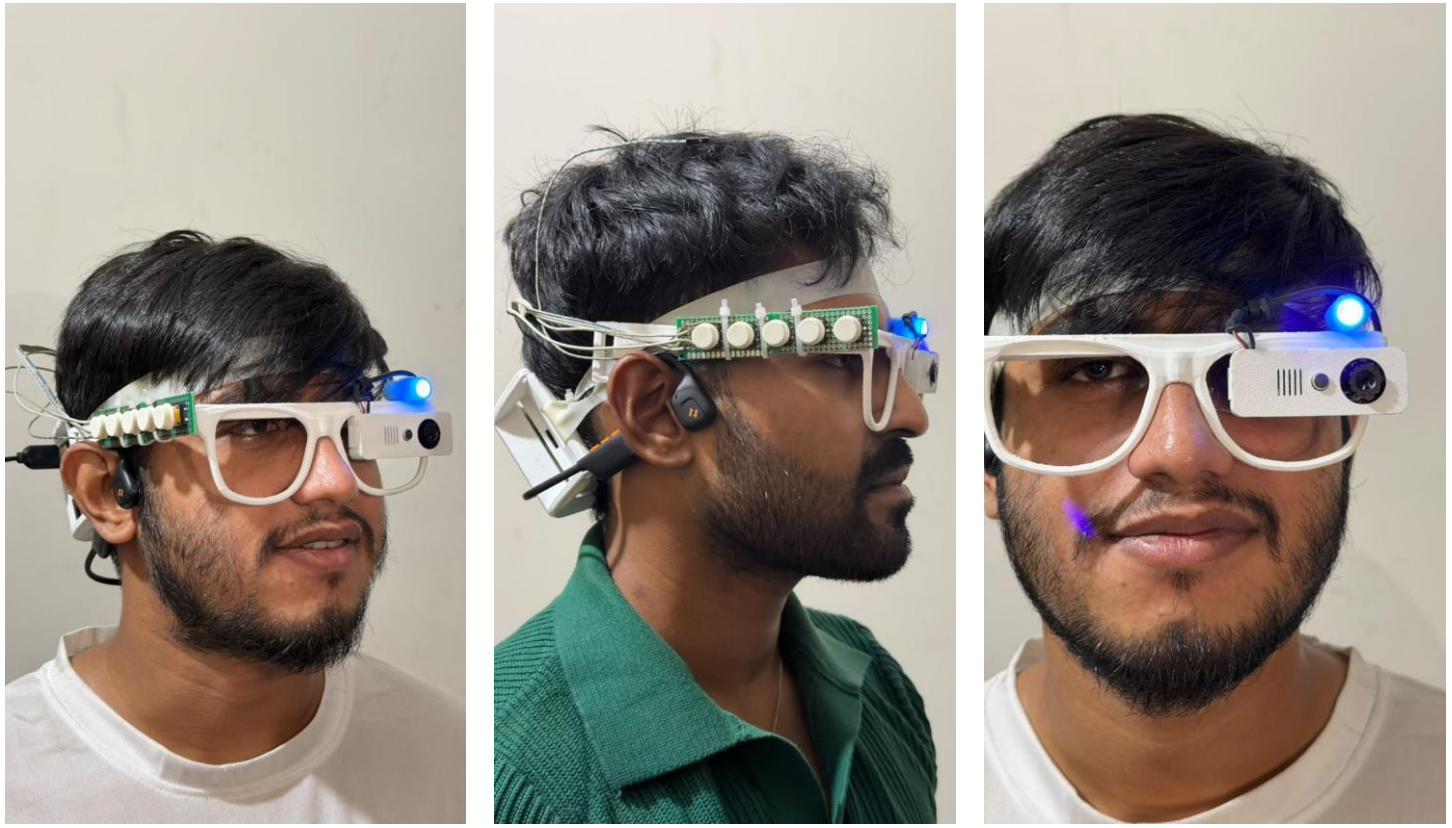
1.48 GB
peak memory, 30-minute run

Heat

- Board idled at 71 °C; peak 88.7 °C, throttled in 76% of samples
- Thermal governor slows background scans when hot
- Text reading is the slowest task

Measured on a Raspberry Pi 5 (8 GB) on 13 October 2026, board at its temperature limit; memory from the 1 October run. Changes after 13 October are not yet timed on the Pi.

Testing the worn glass



Prototype version 2 on team members: fit, weight and button reach are checked before user trials.

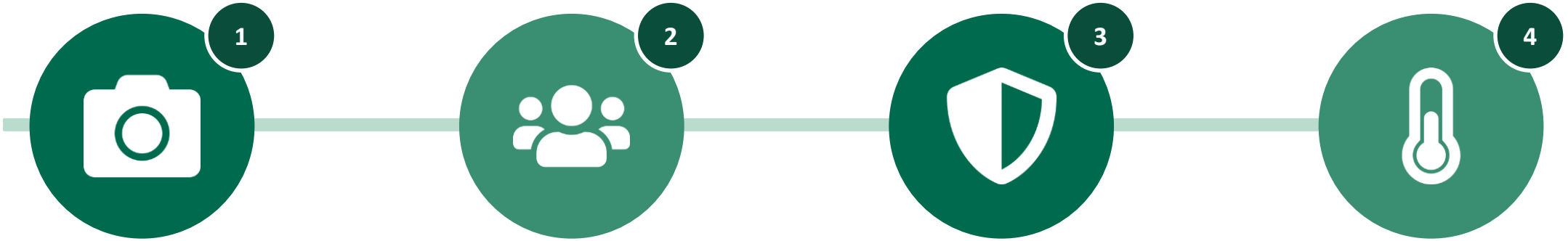
Limitation: the glass camera

In the first field test, currency was unreliable and Bangla text was not read (640×480, fixed focus).

User trials: in progress

Sessions with a visually impaired participant have started. No user study is complete yet.

Limitations and next steps



Camera

Higher resolution or autofocus

User study

8–12 participants, including their own voice commands

More banknotes

Unseen counterfeit prints; UV check on real fakes

Speed and cooling

Time the new code on the Pi; better cooling; battery

Conclusion: from challenges to independence



Illustration

Built

An offline glass that speaks Bangla, reads, names objects and Taka on a Raspberry Pi 5

New methods

Prefix training, a safe verdict, the DMWR watermark check, hand-held fine-tuning

Next

A better camera and the user trials

Thank you

Questions are welcome



References

[1] World Health Organization. Blindness and vision impairment: fact sheet, 2023.

[2] OrCam Technologies. OrCam MyEye.

[3] Ali A., H. et al. A Google Glass based real-time scene analysis for the visually impaired. IEEE Access 9, 2021.

[4] Brilli, D. D. et al. AIris: an AI-powered wearable assistive device for the visually impaired. arXiv:2405.07606, 2024.

[5] Baek, Y. et al. Character region awareness for text detection (CRAFT). CVPR, 2019.

[6] JaidedAI. EasyOCR: ready-to-use OCR with 80+ languages.

[7] Jocher, G., Chaurasia, A., Qiu, J. Ultralytics YOLOv8.

[8] Lin, T.-Y. et al. Microsoft COCO: common objects in context. ECCV, 2014.

[9] Li, S., Deng, W., Du, J. Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild (RAF-DB). CVPR, 2017.

[10] Rhasspy. Piper: a fast, local neural text-to-speech system.

[11] eSpeak NG text-to-speech.

[12] A diverse image dataset for Bangladeshi currency recognition (BanglaTaka). Mendeley Data.

[13] Rafi, M. J. A., Rony, M., Majadi, N. NSTU-BDTAKA: an open dataset for Bangladeshi paper currency detection and recognition. Data in Brief 55, 2024.

[14] Piyas, B. R. C. et al. JaalTaka: a benchmark dataset for detecting counterfeit Bangladeshi banknotes. Mendeley Data V2, 2025.

[15] Piyas, B. R. C. et al. JaalTaka: a benchmark dataset for detecting counterfeit Bangladeshi banknotes. Data in Brief 64, 112377, 2026.

[16] Uddin, M. S., Das, P. P., Roney, M. S. A. Image-based approach for the detection of counterfeit banknotes of Bangladesh. ICIEV, 2016.

[17] Pathak, A. et al. Enhanced counterfeit detection of Bangladesh currency through convolutional neural networks. IJIRCST 12(2), 2024.

[18] Sandler, M. et al. MobileNetV2: inverted residuals and linear bottlenecks. CVPR, 2018.

[19] Chowdhury, A. A. et al. Evaluation of deep learning models on UV ink: a fake money detection scheme with RPN. ICCA (ACM), 2022.

[20] Mutanen, J., Jaaskelainen, T., Parkkinen, J. P. S. Luminescent security properties of banknotes. IS&T PICS Conference, 2003, pp. 421-424.

[21] Nayef, N. et al. ICDAR2019 robust reading challenge on multi-lingual scene text (RRC-MLT-2019). ICDAR, 2019.

[22] Alpha Cephei. Vosk offline speech recognition toolkit.

[23] Piczak, K. J. ESC: dataset for environmental sound classification. ACM Multimedia, 2015.

Links

Our code and results

SARA: glass software for the Raspberry Pi 5
github.com/aflam330/SARA

Research repository: experiments, results, thesis
github.com/aflam330/Senior_project_wearable_glass_for_blind_people_using_ocr_and_ylo

Datasets

JaalTaka (counterfeit notes)
data.mendeley.com/datasets/2m7wk5cy4c/2

BanglaTaka (note photos)
data.mendeley.com/datasets/3cv2sypkkh/1

NSTU-BDTAKA (hand-held notes)
www.sciencedirect.com/science/article/pii/S2352340924006681

Microsoft COCO
cocodataset.org

ICDAR 2019 MLT
rrc.cvc.uab.es/?ch=15

ESC-50 environmental sounds
github.com/karolpiczak/ESC-50

Open-source tools

Ultralytics YOLOv8
github.com/ultralytics/ultralytics

EasyOCR
github.com/JaidedAI/EasyOCR

Piper text-to-speech
github.com/rhasspy/piper

Vosk speech recognition
alphacephei.com/vosk/

eSpeak NG
github.com/espeak-ng/espeak-ng

WHO vision fact sheet
www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment